

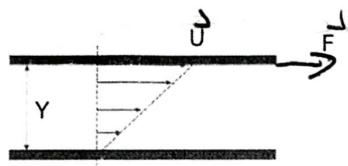
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ChE 311 Quiz 1 Spring 2019

Closed book and closed notes
5 minutes, 5 points

4

1. A type of gear oil was filled between two parallel plate with a gap distance Y , the top plate was moving at constant speed U in x direction under a constant force F . The bottom plate was stationary. Initially, the viscosity of the gear oil was μ . Overtime, the gear oil viscosity was reduced to 0.5μ . Which of the following statement(s) is (are) true.



$$\frac{1}{2} \downarrow \frac{F}{A} = \mu \frac{\partial u}{\partial y} \text{ constant } \downarrow 0.5$$

- A. With the same Y and F , the top plate moved at $2U$.
B. With the same Y and F , the top plate moved at $\frac{1}{2} U$.
C. With the same Y and U , the force is reduced to $\frac{1}{2} F$.
D. None of above.

Answer: C

2. U and V are velocity vectors.

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$$

2.1 $\nabla \cdot \nabla$ is a C.

- (A) vector (B) tensor (C) scalar (D) not enough information to decide

2.2 $\nabla \nabla$ is a B.

- (A) vector (B) tensor (C) scalar (D) not enough information to decide

2.3 $\nabla \cdot \nabla \nabla$ is a A.

- (A) vector (B) tensor (C) scalar (D) not enough information to decide

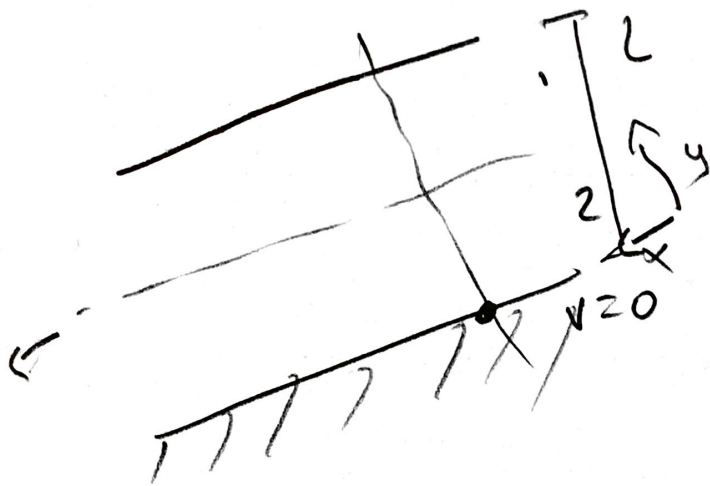
2.4 UV is the same as VU .

- (A) True (B) False

$$\begin{bmatrix} 1 & 4 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \quad \begin{bmatrix} 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Quiz #2

1) Handed-Handed



$$\frac{\partial \zeta}{\partial y} = -\rho g \cos \beta$$

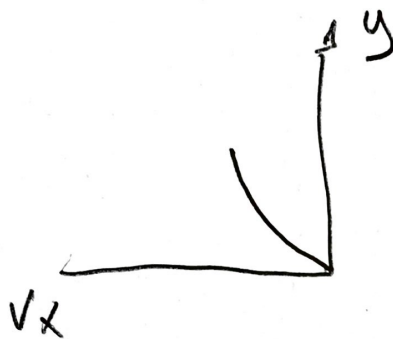
$$\zeta = -\mu \frac{\partial v_x}{\partial y}$$

Need 2 BC

$$\begin{cases} y=0 & v_x=0 \\ y=L & \tau_{xy}=0 \end{cases}$$

$$v_x = \frac{\rho g \cos \theta}{\mu} \left(h - \frac{1}{2} y \right)$$

$$v_x = a - a \left(\frac{y}{L} \right)^2$$



2) →

$$V = v \pi r^2 \quad \left(\frac{1}{2} \right)^2 = 4x$$

$$\zeta = \mu \frac{\partial v}{\partial y} + \frac{1}{2}$$

$$\zeta = ?$$

$$\tau_{xy} = \mu \frac{\partial v}{\partial y}$$

$$\zeta = 8x$$

3)



$$v_x = a \left[1 - \left(\frac{y}{B} \right)^2 \right]$$

$$p_x = -cx$$

$$\begin{aligned} 3. \quad \phi_{xx}|_{x=0, y=0} &= \rho(a(1-0)) \\ &= \rho a \end{aligned}$$

$$1. \quad \phi_{xx}|_{x=0, y=B}$$

$$\begin{aligned} \phi_{xx} &= p + \cancel{\mu \frac{\partial v}{\partial x}} + \rho v_x v_x \\ &= (-\cancel{cx}) + 0 + \rho v_x^2 \\ &= \rho \left(a \left(1 - \left(\frac{B}{B} \right)^2 \right) \right)^2 \\ &= 0 \end{aligned}$$

$$2. \quad \phi_{xy}|_{x=0, y=B}$$

$$\begin{aligned} &= 0 + \tau_{xy} + \rho v_x v_y \\ &= -\mu \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) \\ &= -\mu \frac{\partial}{\partial y} \left[a \left[1 - \left(\frac{y}{B} \right)^2 \right] \right] \\ &= -\mu \left[0 - 2 \frac{y}{B} \right] \end{aligned}$$